

Q10

$$v = f\lambda$$

$$\text{Wavelength } \lambda = 320 / 400 = 0.80 \text{ m}$$

$$\frac{\phi}{2\pi} = \frac{x}{\lambda} \text{ or } \frac{\phi}{2\pi} = \frac{0.20}{0.80}; \quad \phi = \frac{\pi}{2} \text{ rad}$$

Ans: C